

THS8135

TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601) – COMPLIANT FULL SCALE RANGE

SLAS343–MAY 2001

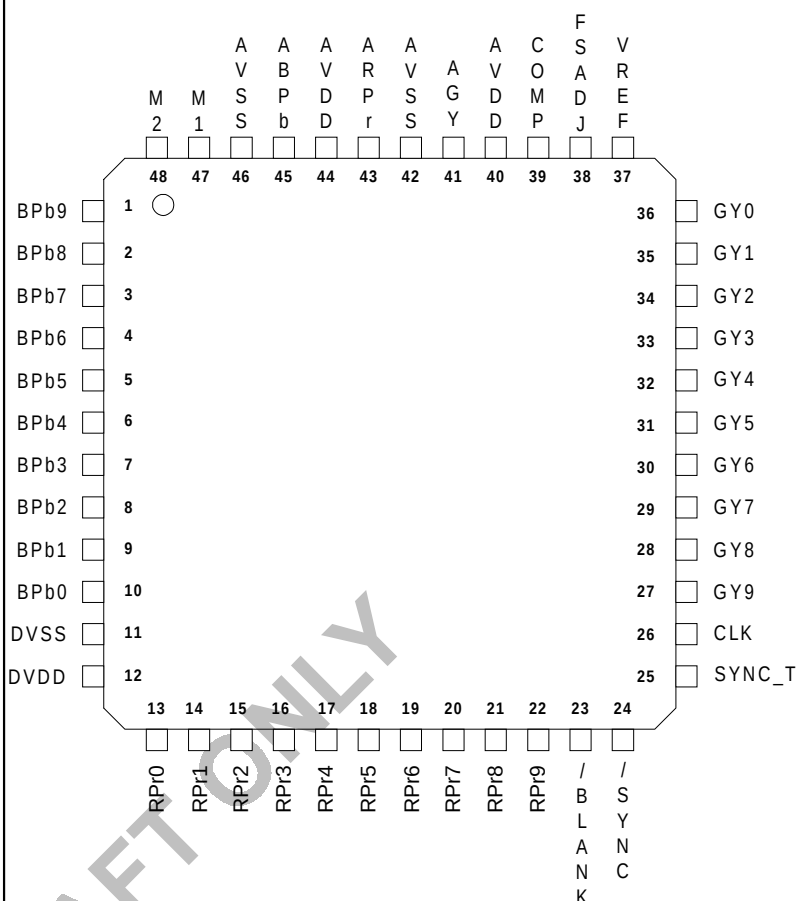
Features

- Triple 10-bit D/A Converters
- 80 (THS8135-80) or 240 MSPS (THS8135-240) Operation
- YUV/RGB Configurable Blanking for Full (0-1023) or Video (ITU-R.BT601) –compliant Input Code Range
- Generic triple DAC mode for non-video applications
- Direct Drive of Doubly-terminated 75Ω Load into Standard Video Levels,irrespective of Operation with Full or Video Input Code Range
- 3x10 bit 4:4:4, 2x10 bit 4:2:2 or 1x10 bit 4:2:2 (ITU-R.BT656) multiplexed YPbPr/GBR input modes
- Bi-level (EIA) or Tri-level (SMPTE) Sync Generation
- Integrated Insertion of Sync-On-Green/Luminance or Sync-On-All Channels
- Internal Voltage Reference

See also: THS8136 (8bit, pin-compatible)

Applications

- High-Definition Television (HDTV) Set-top Boxes / Receivers / Displays
- High-resolution Image Processing



THS8135 is a general-purpose triple high-speed D/A converter optimized for use in video/graphics applications. The device operates from 3.3V analog and 1.8V digital supplies. Its performance is guaranteed at a sampling rate up to 80MSPS (THS8135-80) or 240 MSPS (THS8135-240). THS8135 consists of three 10-bit D/A converters and additional circuitry for bilevel/trilevel sync & blanking level generation. The device can insert a sync on either only the green/luminance ('sync-on-green/'sync-on-Y') channel or on all channels for video applications. A generic triple DAC mode is available making this device suitable for non-video applications as well.

THS8135-80 is a footprint-compatible functional upgrade to THS8133. In addition, THS8135-240 allows a higher update rate for oversampled video digitizing for all PC graphics formats up to UXGA (1600x1200) resolution @ 85Hz and all practical Digital TV formats including HDTV. The support for oversampling significantly reduces the complexity of the analog reconstruction filter required behind the DAC.

Standard video levels can be generated for the full code range (0 to 1023). Alternatively these standard levels can be reached from a reduced code range compliant to the video sampling standard ITU-R.BT-601. In the latter configuration, the full-scale range of the DAC will be dependent on the RGB or YCbCr operation mode of the device, and full video output swing will be reached when driving the device with code range 64-940 for Y,R,G and B channels; and code range 64-960 for Cb&Cr channels.

A digital control input for insertion of a reference (blanking) level on the analog outputs is included. The blanking level will be accurately positioned depending on the RGB/YCbCr operation mode of the device. Irrespective of the operation mode, the inserted sync output amplitude(s) is(are) having the correct ratio to the full-scale video amplitude.

The current-steering DACs can be directly terminated in resistive loads to produce voltage outputs. The device provides a flexible configuration of maximum output current drive. Its output drivers have been specifically designed to produce standard video output levels when directly connected to a single-ended doubly-terminated 75Ω coaxial cable. Finally the input format can be either 3x10 bit 4:4:4, 2x10 bit 4:2:2 or 1x10 bit 4:2:2. This enables a direct interface to a wide range of video DSP/ASIC's including parts generating ITU.R-BT.656 formatted output data.

THS8135

**TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO
(ITU-R.BT601) – COMPLIANT FULL SCALE RANGE**

SLVSXXXX—10/02/00 3:44 PM

Table 1: Available Options

T_A	PACKAGED DEVICES: TQFP-48 PowerPAD™
0°C to 70°C	THS8135-80CPHP
-40°C to 85°C	THS8135-80IPHP
0°C to 70°C	THS8135-240CPHP
-40°C to 85°C	THS8135-240IPHP

Table 2: Terminal description

Name	Pin #		Terminal Description
RPr0-RPr9	13-22	I	Red, Green and Blue resp. Pr, Y and Pb Pixel Data Inputs. Signals with index 0 denote the least significant bits. Refer to functional description for different operating modes.
GY0-GY9	36-27	I	
BPb0-BPb9	10-1	I	
CLK	26	I	Clock input. A rising edge on CLK latches RPr0-9, GY0-9, BPb0-9, /BLANK, /SYNC and SYNC_T. The M1 & M2 inputs are latched by a rising edge on CLK as well but only when additional conditions are satisfied, as explained in its terminal description.
/BLANK	23	I	Blanking Control Input, active low. A rising edge on CLK latches /BLANK. When asserted, the ARPr, AGY and ABPb outputs are driven to the blanking level, irrespective of the value on the data inputs. /SYNC takes precedence over /BLANK so asserting /SYNC (low) while /BLANK is active (low) will result in sync generation. The analog amplitude of the DAC outputs during /BLANK active are determined by the operation mode of the device. /BLANK control is available in both video and generic modes (see functional description).
/SYNC	24	I	Sync Control Input, active low. A rising edge on CLK latches /SYNC. When asserted, only the AGY output (INS3_INT=L, see terminal M2) or ARPr, AGY & ABPb outputs (INS3_INT=H, see terminal M2) are driven to the sync level, irrespective of the values on the data or /BLANK inputs. Consequently, /SYNC should remain low for the whole duration of sync, which is thus in the case of a trilevel sync both the negative and positive portion. See also Figure 6. /SYNC control is only available in video mode (see functional description).
SYNC_T	25	I	Sync Trilevel Control, active high. A rising edge on CLK latches SYNC_T. When asserted, a positive sync (higher than blanking level) is generated when /SYNC is low. When disabled, a negative sync (lower than blanking level) is generated when /SYNC is low. When generating a trilevel (negative-to-positive) sync, a L->H transition on this signal positions the start of the positive transition. See Figure 6 for timing control. When /SYNC is not asserted (high) the value of SYNC_T determines video vs. generic mode of the device (see functional description).

PRODUCT PREVIEW

THS8135**TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601)
– COMPLIANT FULL SCALE RANGE**

SLVSXXXX—10/02/00 3:44 PM

M1	47	I	Operation Mode Control 1. The second rising edge on CLK after a transition on /SYNC latches M1. The interpretation is dependent on the polarity of the last /SYNC transition: /SYNC L->H: latched as M1_INT /SYNC H->L: latched as BLNK_INT. Together with M2_INT, M1_INT configures the device as shown in Table 4. BLNK_INT determines if the device operates with the full or reduced scale range, which also determines the amplitude of the blanking level on the analog output(s) as shown in Table 7.
M2	48	I	Operation Mode Control 2. The second rising edge on CLK after a transition on /SYNC latches M2. The interpretation is dependent on the polarity of the last /SYNC transition: /SYNC L->H: latched as M2_INT /SYNC H->L: latched as INS3_INT Together with M1_INT, M2_INT configures the device as shown in Table 5. When INS3_INT is high, the sync output is inserted on all DAC outputs; a low will insert it only on the AGY output.
ARPr	43	O	Analog Red, Green and Blue resp. Pr, Y and Pb Current Outputs, capable of directly driving a doubly terminated 75 Ω coaxial cable.
AGY	41	O	
ABPb	45	O	
FSADJ	38	I	Full-scale adjust control. The full scale current drive on each of the output channels is determined by the value of a resistor R_{FS} connected between this terminal and AVSS. The nominal value of R_{FS} is TBD Ω . This value produces the nominal full-scale currents of Table 11. The relationship between R_{FS} and the full-scale current level for each operation mode is explained in the functional description.
COMP	39	O	Compensation terminal. A 0.1 μ F capacitor must be connected between COMP and AVDD.
DVDD	12	I	Digital Power Supply (1.8V).
DVSS	11	I	Digital Ground
VREF	37	I/O	Voltage Reference for DACs. An internal voltage reference of nominally 1.2V is provided, which requires an external 0.1 μ F ceramic capacitor between VREF and AVSS. However, the internal reference can be overdriven by an externally supplied reference voltage.
AVDD	40,44	I	Analog Power Supply (3.3V). All AVDD pins must be connected.
AVSS	42,46	I	Analog Ground

PRODUCT PREVIEW

THS8135 TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601) – COMPLIANT FULL SCALE RANGE

SLVSXXXX—10/02/00 3:44 PM

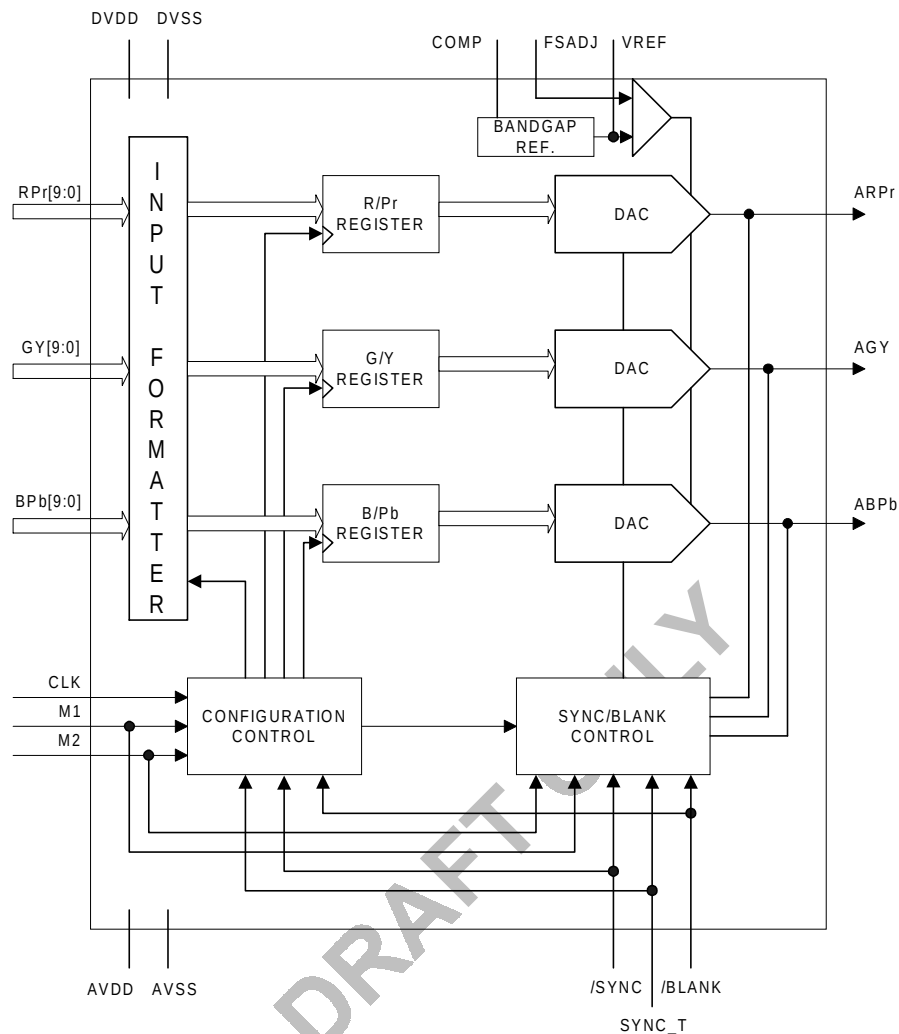


Figure 1: THS8135 block diagram

PRODUCT PREVIEW

Functional Description

THS8135 is a fast well-matched triple DAC with current outputs, optimized for video applications but without sacrificing its usefulness as a general-purpose triple DAC thanks to a 'generic' DAC mode. For video applications, the device can embed an analog output (composite) sync on only the green/luma channel or on all three DAC output channels; offers compatibility with several popular video bus formats and provides standard analog output compliance levels for video digitized according to the ITU-R.BT601 sampling standard. The DAC full-scale range is also adjustable.

Sync generation

Control inputs /SYNC and SYNC_T[rilevel] enable the superposition of an additional current onto the AGY channel or on all three channels, depending on the setting of INS3_INT(see later). Using a combination of the /SYNC and SYNC_T control inputs, either bilevel negative going pulses or trilevel pulses can be generated. Depending on the timing controls for these signals, both horizontal and vertical sync's can be generated in a video system. Assertion of /SYNC (active low) will identify the sync period, while assertion of SYNC_T (active high) within this period will identify the positive excursion of a trilevel sync.

Blanking generation

A control input /BLANK is provided that will fix the output amplitude on all channels to the blanking level, irrespective of the value on the data input ports. The position of the blanking level on each channel and its relation to active video is determined by the operation mode of the device (see later).

Generic mode vs. Video Mode

In video mode, the device provides a dc bias on all channels in RGB mode and on the Y channel in YPbPr mode, according to Figure 7, in order to allow for a negative sync excursion below the blanking level.

Since this introduces a dc bias for the DAC output codes on at least one of the three DAC outputs, this might be undesirable in applications where no sync embedding is needed. Therefore THS8135 includes a 'generic DAC' mode that disables the dc bias on all outputs.

Selection between 'generic' vs. 'video' mode is controlled through a combination of /SYNC, SYNC_T and /BLANK settings. Since in video mode, SYNC_T only determines the sync polarity, this signal has don't care status when no sync insertion takes place i.e. when /SYNC is high. Therefore one of both states can be used for configuring the device in generic mode.

THS8135 enters generic mode when SYNC_T is high outside the sync insertion interval, in other words when both SYNC_T and /SYNC are high. Note that all other features described below (input formatter setting, blank override, color space selection determining blank level position) are still available in generic DAC mode. Generic mode only disables the dc bias, such that the analog DAC output amplitudes corresponding to codes 0 becomes 0V on all channels.

Table 3 shows all combinations of these control signals. Note that when /SYNC is low, it takes precedence over /BLANK.

Table 3: Video vs. generic mode selection

/SYNC	/BLANK	SYNC_T	Operation mode & DAC output
-------	--------	--------	-----------------------------

THS8135

TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601) – COMPLIANT FULL SCALE RANGE

SLVSXXXX—10/02/00 3:44 PM

1	1	1	Generic mode. Blanking override inactive.
1	0	1	Generic mode. Blanking override active. Blanking level position is according to the codes of Table 7, however no dc bias is present on the DAC outputs
1	1	0	Video mode. Blanking override inactive.
1	0	0	Video mode. Blanking override active. Blanking level position is according to the codes of Table 7, with dc bias present on the DAC outputs as shown in Figure 7.
0	X	0	Video mode. Negative sync inserted.
0	X	1	Video mode. Positive sync inserted.

Device Configuration using M1 & M2

The configuration signals M1 and M2 are both¹ sampled on the 2nd rising edge of the CLK input signal after a L->H or H->L transition on /SYNC. Depending on the polarity of this last transition on /SYNC, M1 and M2 are interpreted differently by THS8135, as shown in Table 4.

Table 4: Interpretation of M1

If last event on /SYNC is:	Then M1 is interpreted on 2 nd CLK rising edge following this event as:	Description
H->L	BLNK_INT	Sets operation with Full or Video (ITU-R.BT601)- compliant code Input Range. Selection available in both generic and video modes i.e. the full-scale range is reached from either the 0-1023 code range or the code range of Table 8.
L->H	M1_INT	Sets device operation mode. See Tables 6 & 7.

Table 5: Interpretation of M2

If last event on /SYNC is:	Then M2 is interpreted on 2 nd CLK rising edge following this event as:	Description
H->L	INS3_INT	Sets Sync Insertion mode: /SYNC low enables sync generation on 1 (INS3_INT=L) or all 3 (INS3_INT=H) DAC outputs. SYNC_T determines the sync polarity.
L->H	M2_INT	Sets device operation mode. See Tables 7 & 8.

Selection of Color Space and Input Formatter Mode

¹ In THS8133, only M2 is a sampled signal while M1 is continuously interpreted. By doing so here, the additional input control signal BLNK_INT is generated. See paragraph on backward compatibility with THS8133.

Input data to the device can be supplied from a 3x10b GBR or YPbPr input port. If the device is configured to take data from all three channels, the data is clocked in at each rising edge of CLK. All 3 DAC's operate at the full clock speed of CLK.

In the case of 4:2:2 sampled data (for YPbPr) the device can be fed over either a 2x10 bit or 1x10 bit multiplexed input port. An internal demultiplexer will route input samples to the appropriate DAC: Y at the rate of CLK, Pb & Pr each at rate of $\frac{1}{2}$ CLK.

According to ITU-R.BT-656 the sample sequence is Pb-Y-Pr over a 1x10 bit interface (Y-port). The sample sequence starts at the first rising edge of CLK after /BLANK has been taken high (inactive). Note that in this case the frequency of CLK is 2x the Y conversion speed and 4x the conversion speed of both Pr & Pb.

In the case of a 2x10 bit input interface, both the Y-port and the Pr-port are sampled on every CLK rising edge. The Pr-port carries the sample sequence Pb-Pr. The sample sequence starts at the first rising edge of CLK after /BLANK has been taken high (inactive). Note that in this case the frequency of CLK is equal to the conversion speed of Y and 2x the conversion speed of both Pr & Pb.

Table 6 show the possible configurations of the input formatter, as determined by the internal M1_INT and M2_INT signals. . The color space selection determines also the position of the blanking level, as will be explained next.

Table 6: THS8135 RGB/YCbCr Color Space and Input Formatter Configuration

M1_INT	M2_INT ²	Configuration	Description
L	L	GBR 3x10b- 4:4:4	GBR mode 4:4:4. Data clocked in on each rising edge of CLK from G, B & R input channels.
L	H	YPbPr 3x10b- 4:4:4	YpbPr mode 4:4:4. Data clocked in on each rising edge of CLK from Y, Pb & Pr input channels.
H	L	YPbPr 2x10b- 4:2:2	YPbPr mode 4:2:2 2x10 bit. Data clocked in on each rising edge of CLK from Y channel. A sample sequence of Pb-Pr... should be applied to the Pr port. At the first rising edge of CLK after /BLANK is taken high, Pb should be present on this port.
H	H	YPbPr 1x10b- 4:2:2	YPbPr mode 4:2:2 1x10 bit (ITU-R.BT-656 compliant). Data clocked in on each rising edge of CLK from Y channel. A sample sequence of Pb-Y-Pr should be applied to the Y port. At the first rising edge of CLK after /BLANK is taken high, Pb should be present on this port.

² M1_INT and M2_INT are the internal logic signals set by M1 resp. M2, as explained in Tables 4 & 5.

THS8135

TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601) – COMPLIANT FULL SCALE RANGE

SLVSXXXX—10/02/00 3:44 PM

Selection of Full or Reduced-Scale ITU.BT601 Modes

For each mode, BLNK_INT sets the blanking level generated on the DAC outputs as shown in Table 7, allowing operation with either a full-scale code range or a video code range compliant to ITU-R.BT-601.

Table 7: Full-scale or ITU.BT601 reduced scale mode selection & impact on blanking level positioning

M1_INT	M2_INT	BLNK_INT ³	Operation Mode	Channel Output Level during /BLANK active corresponding to DAC input code: ⁴		
				AGY	ABPb	ARPr
L	L	L	GBR 3x10b 4:4:4, Full Scale Range	0	0	0
L	L	H	GBR 3x10b 4:4:4, ITU-R.BT601-Compliant Range	64	64	64
L	H	L	YPbPr 3x10b 4:4:4, Full Scale Range	0	512	512
L	H	H	YPbPr 3x10b 4:4:4, ITU-R.BT601-Compliant Range	64	512	512
H	L	L	YPbPr 2x10b 4:2:2, ITU-R.BT601-Compliant Range	64	512	512
H	L	H	YPbPr 2x10b 4:2:2, Full Scale Range	0	512	512
H	H	L	YPbPr 1x10b 4:2:2, ITU-R.BT601-Compliant Range	64	512	512
H	H	H	YPbPr 1x10b 4:2:2, Full Scale Range	0	512	512

PRODUCT PREVIEW

In full-scale modes, the DAC will be driven with input codes 0-1023 to the desired video level, set by FSADJ (e.g. a full-scale video amplitude of 700mV when terminated into 37.5 Ohm and when using the nominal R_{FS} value).

In reduced scale ITU-R.BT601 modes, on each of the three channels the same analog output compliance as in full-scale mode will be reached when the DAC is driven with digital inputs within the input code range shown in Table 8. Note that the code range is unequal between RGBY on one hand and CbCr on the other hand in ITU-R.BT601 mode. Not only is the positioning of the blanking level different between both modes, the device also automatically compensates such that the same full-scale video range as in 'full-scale mode' is now reached based on this reduced input code range. Note that even in ITU-R BT.601 mode, the midscale blanking level on Pb and Pr channels is still at code 512 = $[64+(960-64)/2]$. Table 8 shows the valid reduced

³ BLNK_INT is the internal logic signal set by M1, as explained in Table 4.

⁴ Valid for THS8135 in both sync-on-G/Y or sync-on-all configuration modes, and in both generic and video modes.

input code ranges for RGB resp. YCbCr operation on each of the input data busses. If codes outside this range are supplied, they are clipped inside the DAC to the respective min/max.value for the corresponding color component as shown in Table 8. Therefore undershoot/overshoot present in the digital domain will not be detectable on the analog output. Note that while THS8135 allows reduced code range with RGB data, video systems will normally use it only with YCbCr type data.

Table 8: Input code ranges for ITU.BT601 modes

Operating Mode	GY[]	RPr[]	BPb[]
RGB	64-940	64-940	64-940
YCbCr	64-940	64-960	64-960

Internally the device uses digital scaling to accommodate the full 10-bit DAC range from this reduced range. The architecture is optimized to maintain a high linearity and avoids truncation errors in this scaling process.

If the DAC termination is such that the full-scale current corresponds e.g. to a 700mV video output in full-scale mode (as will be the case when the DAC is terminated by 37.5 Ohm and its nominal RFS is applied) for a 1024 input code swing, then in reduced scale mode the resulting full-scale current will also produce a 700mV analog output compliance for a 876(=940-64) code swing on the Y channel and a 896(=960-64) code swing on the Pb and Pr channels.

Finally in all modes, the sync insertion on the analog outputs will be such that a standard-compliant sync amplitude is generated. To guarantee this, the ratio of the DAC's analog full-scale video output (equal in either full or reduced scale modes, as explained above) to the sync amplitude output is fixed and nominally a 7:3 ratio (700mV video vs. 300mV for both negative positive sync excursion amplitude). This ratio is maintained also in reduced scale ITU-R.BT601 mode. Therefore the sync amplitude will maintain its 7:3 ratio, irrespective of either mode, and will also scale proportionally with the video amplitude if FSADJ is changed.

Backward compatibility with THS8133

- Power supply

THS8135 is a functional superset to THS8133 and is footprint compatible i.e. a board designed for THS8133 can also be used with THS8135. Both devices come in the same package and have identical pinouts.

Only the power supply levels need to be adjusted, as shown in Table 9:

Table 9: Power Supply changes THS8133 vs THS8135

	THS8133	THS8135
AVDD	5V	3.3V
DVDD	3.3V or 5V	1.8V

THS8135

TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601) – COMPLIANT FULL SCALE RANGE

SLVSXXXX—10/02/00 3:44 PM

- Device configuration

THS8135 samples both M1 and M2 on the 2nd rising edge of CLK after a transition on /SYNC. Depending on the polarity of the transition, M1 is interpreted as either M1_INT or BLK_INT. In THS8133 the M1 signal is not sampled but continuously interpreted, and is only interpreted as M1_INT. THS8133 does not offer a 'reduced scale' mode and therefore does not require BLNK_INT.

Only when this additional functionality, which is typical for video systems, is desired, a small change in configuration of the device is required by supplying a dynamically changing signal on M1, generated in a similar way as M2, as will be shown in Table 10 below.

Note that this backward compatibility is due to the selection of modes in Table 7: all modes that have equal logic levels for BLNK_INT and M1_INT produce full-scale modes, which are compatible with THS8133. This allows the use of a signal tied high or low on M1, as on THS8133, for the backward compatible full-scale modes.

- DAC outputs

The position of the blanking levels in THS8135 differs from the position of the blanking levels in THS8133. From Figure 7 below, it follows that this device always positions blanking at 350mV, irrespective of the operation mode (YPbPr vs. RGB) or sync insertion selection (sync-on-Y/G or sync-on-all). In THS8133, the position of the blanking level is dependent on the operation mode and sync insertion selection, see THS8133 datasheet. When the DAC output is ac-coupled, as is typically the case, there is no change to the output video waveform. If it is dc-coupled, the consistent blanking level offset in THS8135 makes it easier to level-shift the DAC output signals to the desired range.

- Video vs. generic modes

THS8133 does not offer a generic mode. THS8135 uses only the same number of control signals than THS8133 but additionally introduces a generic video mode by specific use of a don't care signal combination of these control signals on THS8133.

PRODUCT PREVIEW

Programming Example for M2⁵

Configuration of the device will normally be static in a given application although it is theoretically possible to reconfigure the device during operation.

If M2_INT and INS3_INT need to be both low or high, the M2 pin is simply tied low or high. If e.g. M2_INT and INS3_INT need to have different levels, these can be easily derived from the signal on the /SYNC pin, as shown in Table 7 and Figure 2.

Table 10: Generating M2 from /SYNC

In order to have:		Apply to M2:
M2_INT	INS3_INT	
L	H	... /SYNC delayed by 2 CLK periods
H	L	... inverted /SYNC delayed by 2 CLK periods

Similarly M1 can be generated. Therefore at most 1 inverter and 2 flip-flops, are needed to configure any of the THS8135 modes using M1 & M2.

⁵ Programming M1 is analogous.

THS8135
TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO
(ITU-R.BT601) – COMPLIANT FULL SCALE RANGE

SLVSXXXX—10/02/00 3:44 PM

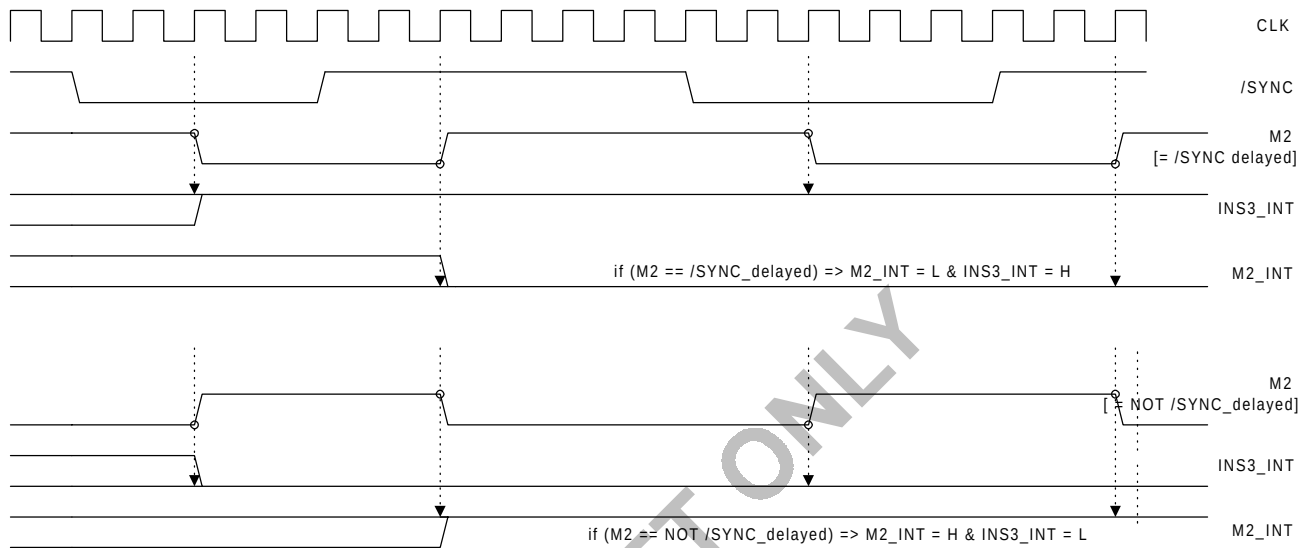
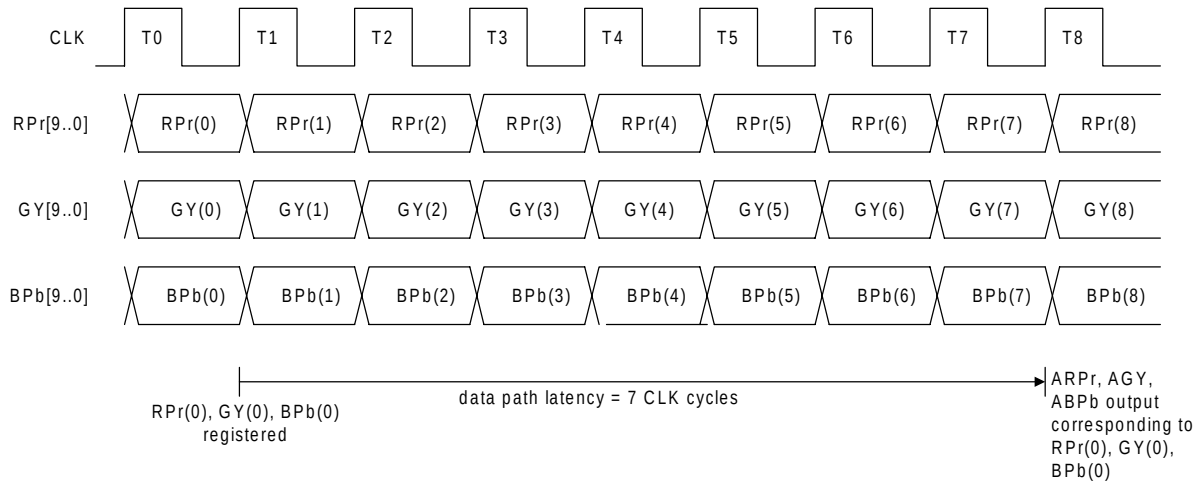
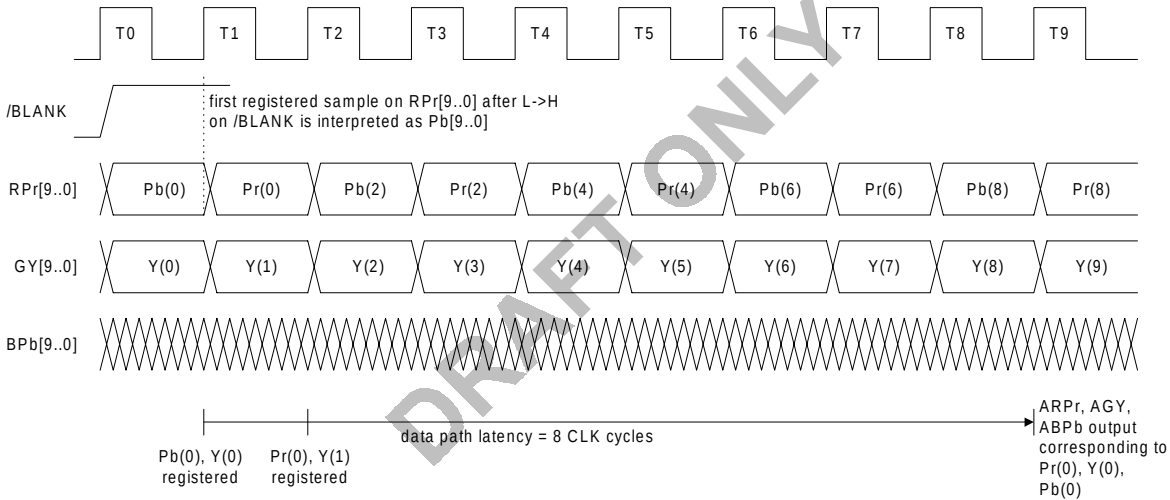
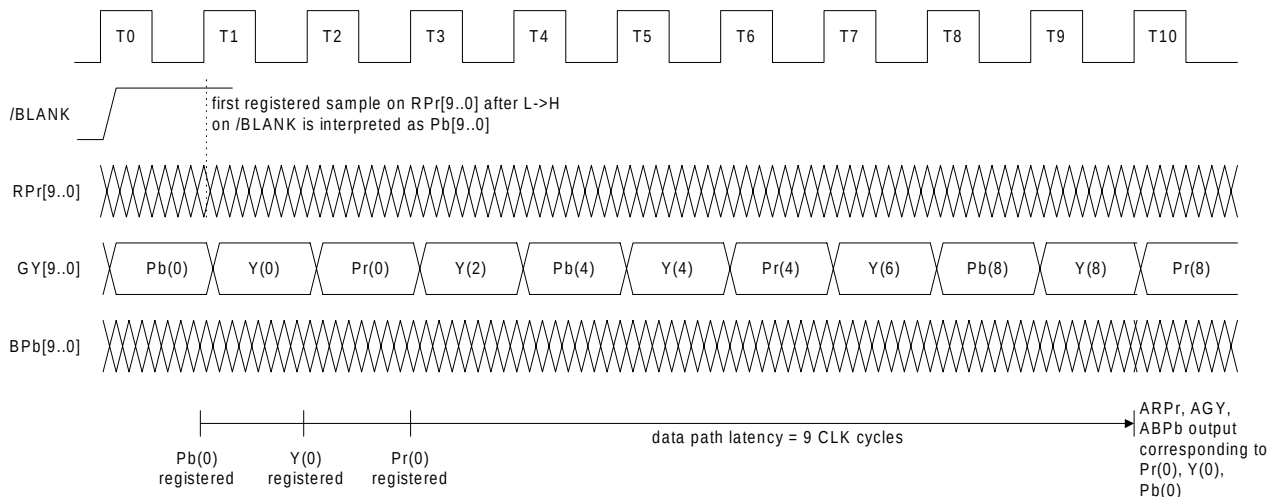


Figure 2: Generating $INS3_INT$ & $M2_INT$ from $M2$

PRODUCT PREVIEW

Figure 3: Input Format & Latency YPbPr 4:4:4 & GBR 4:4:4 Modes**Figure 4: Input Format & Latency YPbPr 4:2:2 2x10bit Mode****Figure 5: Input Format & Latency YPbPr 4:2:2 1x10bit Mode****NOTE: Data path latency numbers on Figs 3-5 are preliminary.**

THS8135 TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601) – COMPLIANT FULL SCALE RANGE

SLVSXXXX—10/02/00 3:44 PM

Figure 6 shows how to control /SYNC, SYNC_T and /BLANK signals to generate trilevel sync levels and blanking at the DAC output in 'video' mode. A bilevel (negative) sync can be generated similarly by avoiding the positive transition on SYNC_T during /SYNC low.

Note in Figure 6 that it is required to keep SYNC_T low outside the sync interval in order to avoid entering 'generic' mode.

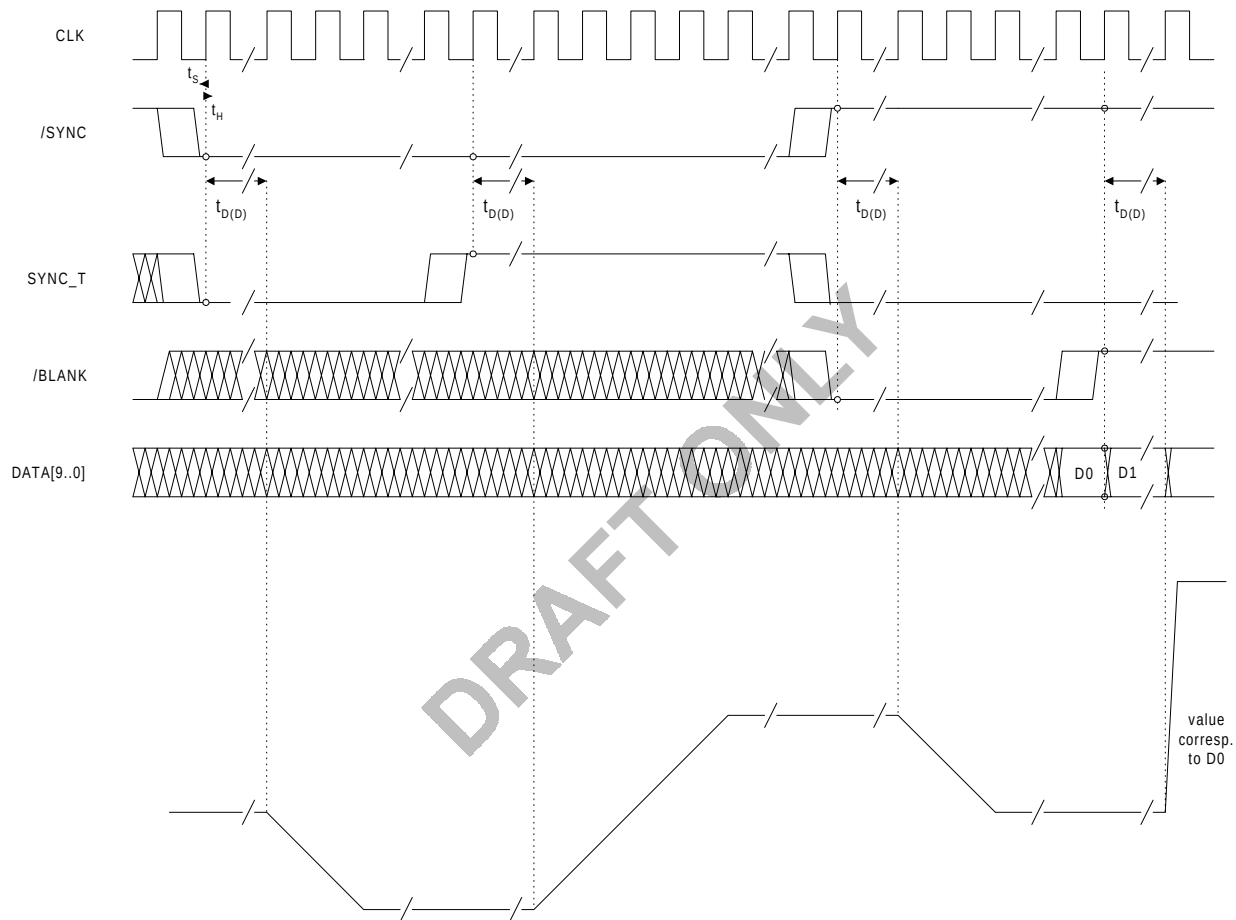


Figure 6: Sync & blanking generation

PRODUCT PREVIEW

DAC operation

The analog output drivers generate a current of which the drive level can be user-modified by choice of an appropriate resistor value R_{FS} , connected to the FSADJ pin. Refer to the paragraph on output amplitude control for details on how the output drive is affected by the operation mode of the device.

All current sources derive their amplitudes from an internal generator that produces a 1.2V reference level. All current source amplitudes (video, blanking, sync) are derived from this reference so that the relative amplitudes of sync/blank/video are always equal to their nominal relationships. For increased stability on the absolute levels, the user can overdrive the reference by directly driving the VREF terminal.

Figure 7 shows the nominal voltage output levels in video mode. Note that in all modes (RGB & YPbPr) the blanking level corresponds to 350mV. The position of the blanking level stays constant in both sync-on-Y or sync-on-all modes, and whether actual sync insertion is used or not on the device. In generic mode, the DAC output swing for full-scale or reduced-scale input codes (depending on device configuration) is 0 to 700mV.

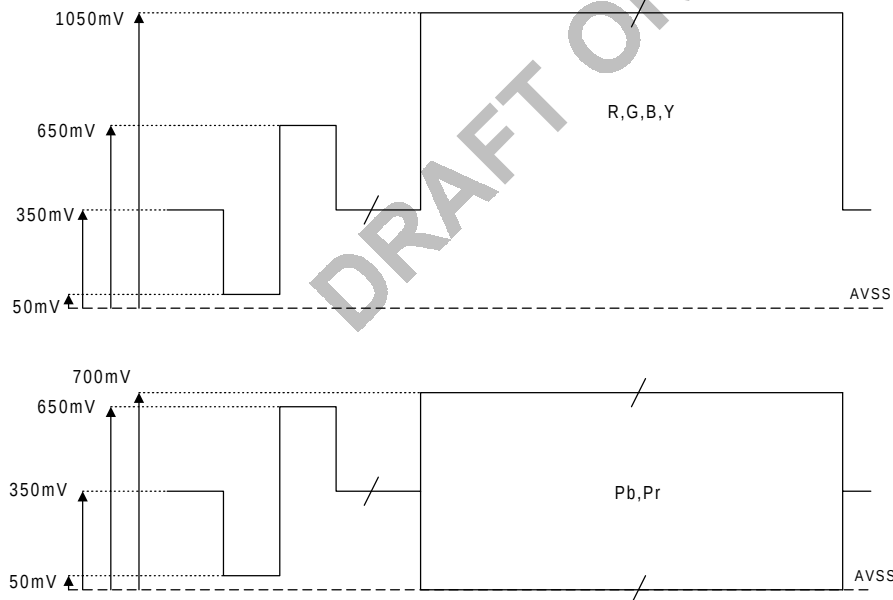


Figure 7: Typical DAC output waveforms in 'video' mode

THS8135

**TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO
(ITU-R.BT601) – COMPLIANT FULL SCALE RANGE**

SLVSXXXX—10/02/00 3:44 PM

Output amplitude control

The current drive on all three output channels and on the internal sync generator is controlled by a resistor R_{FS} that must be connected between FSADJ and AVSS. In all operation modes the relative amplitudes of the current drivers are maintained irrespective of the R_{FS} value, as long as a maximum current drive capability is not exceeded.

The sync generator is composed of different current sources that are internally routed to a corresponding DAC output. Depending on the setting of INS3_INT during /SYNC low, the sync current drive is added to either only the green channel output (sync-on-green) if INS3_INT=L or all three channel outputs if INS3_INT=H, when in video mode. In either case the relative current levels, as defined below, are maintained.

The exact relationship between R_{FS} and the current drive level on each channel is dependent on the operation mode of the device: YCbCr vs RGB and 'full-scale mode' or 'reduced ITU-R.BT601 mode'.

The device has an internal voltage reference derived from a bandgap reference of 1.2V. The relationship between the full-scale current drive level (in full-scale mode) and R_{FS} is given by:

$$I_{FS} [\text{mA}] = \alpha \times V_{REF} [\text{V}] / R_{FS} [\Omega]$$

where α is dependent on the operation mode of the device.

DRAFT ONLY

PRODUCT PREVIEW

THS8135
TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601)
– COMPLIANT FULL SCALE RANGE

SLVSXXXX—10/02/00 3:44 PM

Typical full-scale currents are shown in Table 11. These are reached for the nominal R_{FS} value. When terminated with the nominal resistive load, voltages of standard video levels will be reached. The user is free to connect another resistor value, but care should be taken not to exceed the maximum current level on each of the DAC outputs, as shown in the Specifications section.

Table 11 shows the nominal full-scale current for all 'full-scale modes'. If the device is configured in one of the 'ITU-R.BT601 reduced scale' modes, the same full-scale current as shown in Table 11 will be produced for the reduced input code swing as defined in Table 8.

Table 11: THS8133 nominal full-scale currents

Operation Mode Description	AGY		ARPr		ABPb	
	I_{FS} (mA)	α	I_{FS} (mA)	α	I_{FS} (mA)	α
GBR	28 ⁶	TBD	28	TBD	28	TBD
YPbPr	28	TBD	18.67 ⁸	TBD	18.67	TBD

⁶ $I_{FS} = 1050\text{mV}/37.5\Omega$

⁸ $I_{FS} = 700\text{mV}/37.5\Omega$

THS8135

TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601) – COMPLIANT FULL SCALE RANGE

SLVSXXXX—10/02/00 3:44 PM

Specifications

Absolute maximum ratings over operating free-air temperature range

Supply voltage: AV_{DD} to AV_{SS}	–0.5V to 3.6V
DV_{DD} to DV_{SS}	–0.5V to 2.0V
Supply voltage: AV_{DD} to DV_{DD} , AV_{SS} to DV_{SS}	–0.5V to 0.5V
Digital input voltage range to DV_{SS}	–0.5V to $DV_{DD}+0.5V$
Operating free-air temperature range, T_A	0°C to 70°C (C grade) –40°C to 85°C (I grade)
Storage temperature range, T_{STG}	–55°C to 150°C

Stresses beyond those listed under ‘absolute maximum ratings’ may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under ‘recommended operating conditions’ is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

Recommended operating conditions over operating free-air temperature range T_A

Power supply

		MIN	NOM	MAX	UNIT
Supply voltage	AV_{DD}	3.0	3.3	3.6	V
	DV_{DD}	1.65	1.8	2.0	

Digital & reference inputs

		MIN	NOM	MAX	UNIT
V_{IH}	High-level input voltage	1.35		DV_{DD}	V
V_{IL}	Low-level input voltage	DV_{SS}		0.8	V
f_{clk}	Clock frequency	0		80/240	MHz
$t_{w(CLKH)}$	Pulse duration, clock high	TBD			ns
$t_{w(CLKL)}$	Pulse duration, clock low	TBD			ns
V_{REFI}	Reference input voltage ⁹ , see note 1		1.2	TBD	V

⁹ Voltage reference input applies to the externally applied voltage (overdrive condition). Internally a 2KΩ resistor isolates the internal reference from the externally applied voltage, if any.

THS8135
TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601)
– COMPLIANT FULL SCALE RANGE

SLVSXXXX—10/02/00 3:44 PM

R_{FS}	FSADJ resistor, see note 1	TBD	TBD		Ω
----------	-------------------------------	-----	-----	--	----------

Notes

1. The combination of V_{REF} and R_{FS} can be chosen at will as long as the maximum full-scale DAC output current I_{FS} does not exceed TBD% of its nominal value. Therefore, at fixed $R_{FS} = R_{FS(nom)}$, V_{REF} should not be higher than the max. value mentioned and at fixed $V_{REF} = V_{REF(nom)}$, R_{FS} should not be less than the min. value mentioned.

DRAFT ONLY

PRODUCT PREVIEW

THS8135

**TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO
(ITU-R.BT601) – COMPLIANT FULL SCALE RANGE**

SLVSXXXX—10/02/00 3:44 PM

Electrical characteristics over recommended operating conditions with $f_{CLK} = 80/240$ MSPS and use of internal reference voltage V_{REF} (unless otherwise noted), with $R_{FS} = R_{FS(nom)}$

Power supply (1 MHz, -1dBFS digital sine simultaneously applied to all 3 channels)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
I_{DD}	Operating supply current	$AV_{DD} = 3.3\text{ V}$, $DV_{DD} = 1.8\text{ V}$		TBD	TBD	mA
P_D	Power dissipation	$AV_{DD} = 3.3\text{ V}$ $DV_{DD} = 1.8\text{ V}$		TBD	TBD	mW

Digital inputs – DC characteristics

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
I_{IH}	High-level input current	$AV_{DD} = 3.3\text{ V}$			+1	uA
I_{IL}	Low-level input current	$DV_{DD} = 1.8\text{ V}$			-1	uA
$I_{IL(CLK)}$	Low-level input current, CLK & $I_{IH(CLK)}$ High-level input current, CLK	Digital inputs & CLK at 0 V for I_{IL} ; digital inputs & CLK at 2.0V for I_{IH}	-1		+1	uA
C_i	input capacitance	$T_A = 25^\circ\text{C}$		5		pF
t_s	Data & Control inputs setup time		1			ns
t_h	Data & Control inputs hold time		0			ns
$t_{D(D)}$	Digital process delay ¹⁰ from first registered color component of pixel	RGB & YpbPr 4:4:4		TBD		CLK periods
		YPbPr 4:2:2 2x10 bit		TBD		
		YPbPr 4:2:2 1x10 bit		TBD		

¹⁰ This parameter is guaranteed by design. The digital process delay is defined as the number of CLK cycles required for the first registered color component of a pixel, starting from the time of registering it on the input bus, to propagate through all processing and appear at the DAC output drivers. The remaining delay through the IC is the analog delay $t_{D(A)}$ of the analog output drivers.

THS8135
**TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601)
– COMPLIANT FULL SCALE RANGE**

SLVSXXXX—10/02/00 3:44 PM

Analog (DAC) outputs

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
DAC resolution				10	10		bits
INL	Integral Nonlinearity	Static, best-fit			+/- 0.6	+/- 1.2	LSB
DNL	Differential Nonlinearity	Static			-0.25 / +0.5	+/- 1	LSB
PSRR	Power Supply Ripple Rejection Ratio of DAC Output (Full Scale)	Note 1.		40			dB
		f = DC to 100 kHz f = 100 kHz to 1 MHz		15			dB
XTALK	Crosstalk between channels	f up to 30 MHz. Note 2.				TBD	dB
V _{refo}	Voltage Reference Output			TBD	1.2	TBD	V
R _R	VREF Output Resistance			TBD	TBD	TBD	Ω
K _{DAC}	DAC gain factor				see table 9		
K _{IMBAL}	Imbalance between DACs	Note 3.				+/-2%	
V _{OC}	DAC Output Compliance Voltage (video only)	R _L = 37.5 Ω, Note 4.	Video mode		0.7	TBD	V
			Generic mode			1.2	
I _{FS}	GBR	Int.	AGY	TBD	28	TBD	mA
		Ref.	ABPb & ARPr	TBD	28	TBD	
		Ext.	AGY	TBD	28	TBD	
		Ref.	ABPb & ARPr	TBD	28	TBD	
	YPbPr	Int.	AGY	TBD	28	TBD	
		Ref.	ABPb & ARPr	TBD	18.67	TBD	
		Ext.	AGY	TBD	28	TBD	

PRODUCT PREVIEW information concerns products in the formative or design phase of development. Characteristic data and other specifications are design goals. Texas Instruments reserves the right to change or discontinue these products



Copyright © 2000 Texas Instruments Incorporated

PRODUCT PREVIEW

THS8135

**TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO
(ITU-R.BT601) – COMPLIANT FULL SCALE RANGE**

SLVSXXXX—10/02/00 3:44 PM

			ABPb & ARPr	TBD	18.67	TBD	
R_{OUT}	DAC Output Resistance			TBD	TBD		Ω
C_{OUT}	DAC Output Capacitance (pin capacitance)				5		pF
t_{RDAC}	DAC Output Current Risettime	10 to 90% of full-scale			2		ns
t_{FDAC}	DAC Output Current Faltime	10 to 90% of full-scale			2		ns
$t_{D(A)}$	Analog Output Delay	Note 5. Measured from $CLK=V_{IH(min)}$ to 50% of full-scale transition				TBD	ns
t_s	Analog Output Settling Time	Note 6. measured from 50% of full scale transition on output to output settling, within 2%			TBD	3	ns
SNR	Signal to Noise Ratio	1MHz, -1dBFS digital sine input			57.5		dB
SFDR	Spurious-free Dynamic Range	1MHz, -1dBFS digital sine input			64		dB
BW(1dB)	Bandwidth				40 (THS8135-80) 120 (THS8135-240)		MHz
E_{glitch}	Glitch energy	full-scale code transition @ 240 MSPS			25		pVs

PRODUCT PREVIEW

Notes:

1. PSRR is measured with a 0.1uF capacitor between the COMP and AVDD pin; with a 0.1uF capacitor connected between the VREF pin and AVSS. The ripple amplitude is within the range 100mVp-p to 500mVp-p with the DAC output set to full scale and a double-terminated 75 Ω (=37.5 Ω) load. PSRR is defined as $20 \cdot \log(\text{ripple voltage at DAC output} / \text{ripple voltage at AVDD input})$. Limits from characterization only.
2. Crosstalk spec applies to each possible pair of the 3 DAC outputs. Limit from characterization only.
3. The imbalance between DACs applies to all possible pairs of the three DACs.

THS8135

TRIPLE 10-BIT, 80/240 MSPS VIDEO DAC WITH TRI-LEVEL SYNC AND VIDEO (ITU-R.BT601) – COMPLIANT FULL SCALE RANGE

SLVSXXXX—10/02/00 3:44 PM

4. Nominal values at $R_{FS}=R_{FS(nom)}$; Maximum values at $R_{FS}=TBD R_{FS(nom)}$. Maximum limits from characterization only.
5. This value excludes the digital process delay, $t_{D(D)}$. Limit from characterization only.
6. Maximum limit from characterization only.

DRAFT ONLY

PRODUCT PREVIEW

IMPORTANT NOTICE

Texas Instruments and its subsidiaries (TI) reserve the right to make changes to their products or to discontinue any product or service without notice, and advise customers to obtain the latest version of relevant information to verify, before placing orders, that information being relied on is current and complete. All products are sold subject to the terms and conditions of sale supplied at the time of order acknowledgment, including those pertaining to warranty, patent infringement, and limitation of liability.

TI warrants performance of its products to the specifications applicable at the time of sale in accordance with TI's standard warranty. Testing and other quality control techniques are utilized to the extent TI deems necessary to support this warranty. Specific testing of all parameters of each device is not necessarily performed, except those mandated by government requirements.

Customers are responsible for their applications using TI components.

In order to minimize risks associated with the customer's applications, adequate design and operating safeguards must be provided by the customer to minimize inherent or procedural hazards.

TI assumes no liability for applications assistance or customer product design. TI does not warrant or represent that any license, either express or implied, is granted under any patent right, copyright, mask work right, or other intellectual property right of TI covering or relating to any combination, machine, or process in which such products or services might be or are used. TI's publication of information regarding any third party's products or services does not constitute TI's approval, license, warranty or endorsement thereof.

Reproduction of information in TI data books or data sheets is permissible only if reproduction is without alteration and is accompanied by all associated warranties, conditions, limitations and notices. Representation or reproduction of this information with alteration voids all warranties provided for an associated TI product or service, is an unfair and deceptive business practice, and TI is not responsible nor liable for any such use.

Resale of TI's products or services with statements different from or beyond the parameters stated by TI for that product or service voids all express and any implied warranties for the associated TI product or service, is an unfair and deceptive business practice, and TI is not responsible nor liable for any such use.

Also see: Standard Terms and Conditions of Sale for Semiconductor Products. www.ti.com/sc/docs/stdterms.htm

Mailing Address:

Texas Instruments
Post Office Box 655303
Dallas, Texas 75265